Does the distance to rho move with $\log_2 r$? (priced DP residual collectors)

Pricing caveat (from the independent validator, added after validation). In the band as built, `residual_table_bytes_uncharged` is a *byte* count (table entries × (16 + 8·columns)), but its `_uncharged` name makes the band price it as *operations* at weight w. For the baseline collector it is 86–92% of priced cost at w = 1 and 52–69% at w = 0.1; the candidates store only distinguished points, so their counter is about 150–210× smaller. **The large w > 0 candidate/baseline gains and the 48–53× baseline distances to rho in this report are therefore mostly an artifact of that pricing, not operation-count savings.** The w = 0 results are unaffected. The tables below put the band as built beside the band with that one counter excluded (same data, same pairing).

curve	ratio (w = 0 / 0.1 / 1)	as built	byte counter excluded
p36	res1 → res1eq	0.53 / 0.29 / 0.13	0.53 / 0.60 / 0.90
p36	res4 → res4eq	0.67 / 0.24 / 0.09	0.67 / 0.76 / 1.13
p36	res1 → res1dp	0.87 / 0.42 / 0.13	0.87 / 0.87 / 0.87
p36	res4 → res4dp	0.92 / 0.29 / 0.08	0.92 / 0.92 / 0.92
p36	rho → res4	3.70 / 11.77 / 48.09	3.70 / 3.70 / 3.70
p36	rho → res1eq	1.32 / 1.49 / 2.27	1.32 / 1.48 / 2.21
p36	rho → res1dp	2.15 / 2.17 / 2.25	2.15 / 2.15 / 2.15
p40	res1 → res1eq	0.68 / 0.37 / 0.17	0.68 / 0.77 / 1.18
p40	res4 → res4eq	0.63 / 0.22 / 0.08	0.63 / 0.71 / 1.09
p40	res1 → res1dp	0.90 / 0.43 / 0.13	0.90 / 0.90 / 0.90
p40	res4 → res4dp	0.84 / 0.26 / 0.07	0.84 / 0.84 / 0.84
p40	rho → res4	4.10 / 13.04 / 53.28	4.10 / 4.10 / 4.10
p40	rho → res1eq	1.72 / 1.95 / 2.99	1.72 / 1.95 / 2.98
p40	rho → res1dp	2.27 / 2.27 / 2.28	2.27 / 2.27 / 2.27
p42	res1 → res1eq	0.55 / 0.30 / 0.14	0.55 / 0.62 / 0.95
p42	res4 → res4eq	0.32 / 0.18 / 0.08	0.32 / 0.36 / 0.55
p42	res1 → res1dp	0.76 / 0.37 / 0.11	0.76 / 0.76 / 0.76
p42	res4 → res4dp	0.44 / 0.22 / 0.07	0.44 / 0.44 / 0.44
p42	rho → res4	6.76 / 13.76 / 45.27	6.76 / 6.76 / 6.76
p42	rho → res1eq	1.39 / 1.58 / 2.42	1.39 / 1.58 / 2.42
p42	rho → res1dp	1.92 / 1.92 / 1.93	1.92 / 1.92 / 1.92

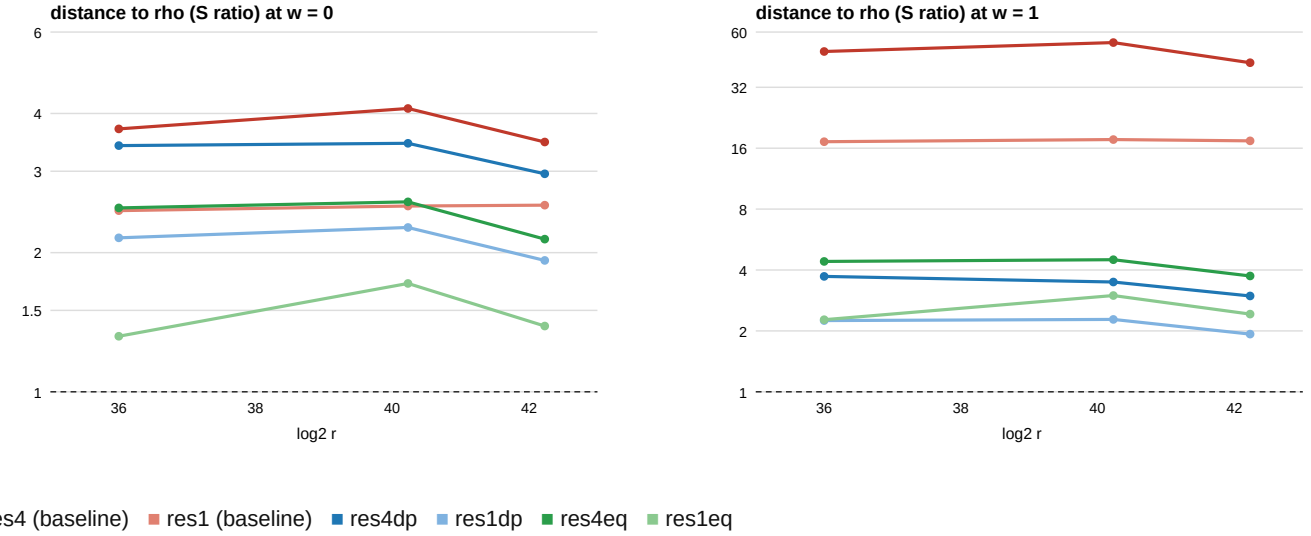
Reading with the byte counter excluded: **dp** is a constant 0.76–0.94× the baseline at every weight (a modest real saving; 0.44 for res4 on p42, whose baseline has one censored run); **eq** is 0.54–0.68× at w = 0 but 0.9–1.18× at w = 1 (0.55 for res4 on p42, same censored baseline) (it pays about 2.5 priced operations per step, so it is no better than the baseline at w = 1 once bytes are not counted). The distance to rho of the baseline collapses to 2.5–4.1× at every weight (res1 ≈ 2.5, res4 ≈ 3.4–4.1). Rho's own counters are also a lower bound (cycle-window scan, jump index and DP test not counted), which makes w > 0 distances to rho look larger than they are. Whether bytes should be priced as operations is the owner's call (§5a); nothing here changes the band.

What was measured. The modes of `research/ic_residual_dp_eq_20261101` (`residual-fused-dp`, `residual-fused-eq`, same per-step pricing) on three prime-order curves: p36 ($2^{36.0}$) from the earlier studies, and p40 ($2^{40.2}$) and p42 ($2^{42.2}$) generated by `scripts/gen_curve.py`. 12 targets × 2 rounds per curve, B = 1 and 4, counted

unit, 864 records, 863 verified and one baseline run exhausted (below), `ecbench verify --replay 12` audits OK, A/A = 1.0000. Local L0 run, counts only.

Distance remaining to matched rho (S ratio rho → arm: w = 0 / 0.1 / 1)

arm	p36	p40	p42
res4 (baseline)	3.70 / 11.77 / 48.09	4.10 / 13.04 / 53.28	3.47 / 10.54 / 42.35
res1 (baseline)	2.46 / 5.15 / 17.22	2.52 / 5.27 / 17.65	2.53 / 5.24 / 17.40
res4dp	3.41 / 3.47 / 3.72	3.45 / 3.45 / 3.49	2.96 / 2.97 / 2.98
res1dp	2.15 / 2.17 / 2.25	2.27 / 2.27 / 2.28	1.92 / 1.92 / 1.93
res4eq	2.50 / 2.85 / 4.41	2.58 / 2.92 / 4.50	2.14 / 2.43 / 3.74
res1eq	1.32 / 1.49 / 2.27	1.72 / 1.95 / 2.99	1.39 / 1.58 / 2.42



From 2^{36} to 2^{42} the distance does not grow: res1eq 1.32 → 1.72 → 1.39× rho at w = 0 and 2.27 → 2.99 → 2.42× at w = 1; res1dp 2.15 → 2.27 → 1.92× at w = 0; the baseline res4 3.70 → 4.10 → 3.47× at w = 0 but 48 → 53 → 42× at w = 1. The collector keeps its fixed overhead (√(B + 1) collisions plus per-step priced work) at every size measured; there is no sign of it closing on rho or falling behind with r over this range. Three sizes and 12 targets each cannot fix a slope.

Counted S ratio, candidate / baseline (paired; 95% interval; green = below 1)

curve	pair	w = 0	w = 0.1	w = 1
p40	res1dp/res1	0.898 [0.683, 1.182]	0.430 [0.327, 0.566]	0.129 [0.098, 0.170]
p40	res1eq/res1	0.680 [0.495, 0.935]	0.369 [0.269, 0.508]	0.169 [0.123, 0.233]
p40	res4dp/res4	0.841 [0.742, 0.953]	0.265 [0.234, 0.300]	0.065 [0.058, 0.074]
p40	res4eq/res4	0.628 [0.533, 0.740]	0.224 [0.190, 0.264]	0.084 [0.072, 0.100]
p42	res1dp/res1	0.759 [0.601, 0.959]	0.368 [0.293, 0.461]	0.111 [0.089, 0.138]
p42	res1eq/res1	0.548 [0.444, 0.675]	0.301 [0.247, 0.367]	0.139 [0.115, 0.169]
p42	res4dp/res4	0.438 [0.148, 1.295]	0.285 [0.251, 0.323]	0.071 [0.063, 0.081]
p42	res4eq/res4	0.316 [0.110, 0.914]	0.233 [0.212, 0.255]	0.089 [0.082, 0.097]
p36	res1dp/res1	0.873 [0.695, 1.097]	0.422 [0.336, 0.530]	0.131 [0.104, 0.164]
p36	res1eq/res1	0.535 [0.423, 0.677]	0.290 [0.229, 0.367]	0.132 [0.104, 0.168]
p36	res4dp/res4	0.920 [0.782, 1.083]	0.295 [0.250, 0.346]	0.077 [0.066, 0.091]

p36	res4eq/res4	0.675 [0.595, 0.764]	0.242 [0.213, 0.274]	0.092 [0.081, 0.104]

Frozen rule C1 (p40 and p42, 8 cells, below 1 at w = 0, 0.1 and 1): **not met**, 6 of 8 cells hold. The failures are both at w = 0: p40 res1dp [0.683, 1.182] (12 targets, wide) and p42 res4dp [0.148, 1.295]. The latter comes from one **censored baseline run**: res4 on p42, round 1, was `exhausted` at 10^8 trials (S = 87.9 recorded). The baseline stores every residual in a table capped at 2^{22} entries (`TABLE_CAP`); at 2^{42} it needs about $8{\cdot}10^6$ steps, so after the cap later steps can only collide with the first $4.2{\cdot}10^6$ residuals, collisions become rarer, and this run hit the trial limit. The candidates store only distinguished points and stay far below the cap. All w = 0.1 and w = 1 cells hold (0.22–0.43 and 0.065–0.17).

Classification and caveats

- Engineering of a generic collision search, as before; rho stays ahead at every weight (1.3–3.0× for the best arms), no structural index-calculus content, no claim to beat rho.
- The baseline's table cap is a limit of the existing collector, not of the candidates, and it affects only the baseline arms at large r; the censored pair is reported, not dropped.
- Curves p40 and p42 were generated for this study (random curve with a prime group order, found by a Hasse-interval BSGS); gen_curve.py checks that N is prime, lies in the Hasse interval and that N·P = O for the generator P, which fixes the group order. Distinct seeds from earlier studies, so sessions are not pooled.
- No `VALIDATION.json`; no leaderboard or benchmarks row. Isolation L0; L2 specs committed for an isolab-eks replay.